EXPERIMENT 1 – ROOM REVERBERATION USING LTI SYSTEMS

1. Objective

To simulate room reverberation by introducing delayed and attenuated reflections into an audio signal and study how the reflections affect the output.

2. Approach

Impulse Response Construction

- We constructed an impulse response consisting of an initial unit impulse representing the direct sound, followed by delayed impulses representing room reflections.
- We considered two patterns for the amplitudes of the reflections : 2^(-i) and 3^(-i).
- Delays in seconds were converted into samples using the sampling frequency. Zeros were maintained between impulses to represent the time gaps between reflections.

Convolution and Audio Processing

- We implemented linear convolution manually to combine the input audio with the constructed impulse response.
- Each reflection produces a delayed and scaled copy of the original signal. These copies are added to obtain the reverberated output.
- The same impulse response was applied independently to the left and right channels of the stereo recording.

Parameter Variation

We studied the effect of changing two parameters:

- **Number of reflections:** 3, 5 and 8 reflections, keeping the spacing at 0.10 s.
- **Delay spacing:** 0.05 s, 0.10 s and 0.20 s, keeping the number of reflections at 5.

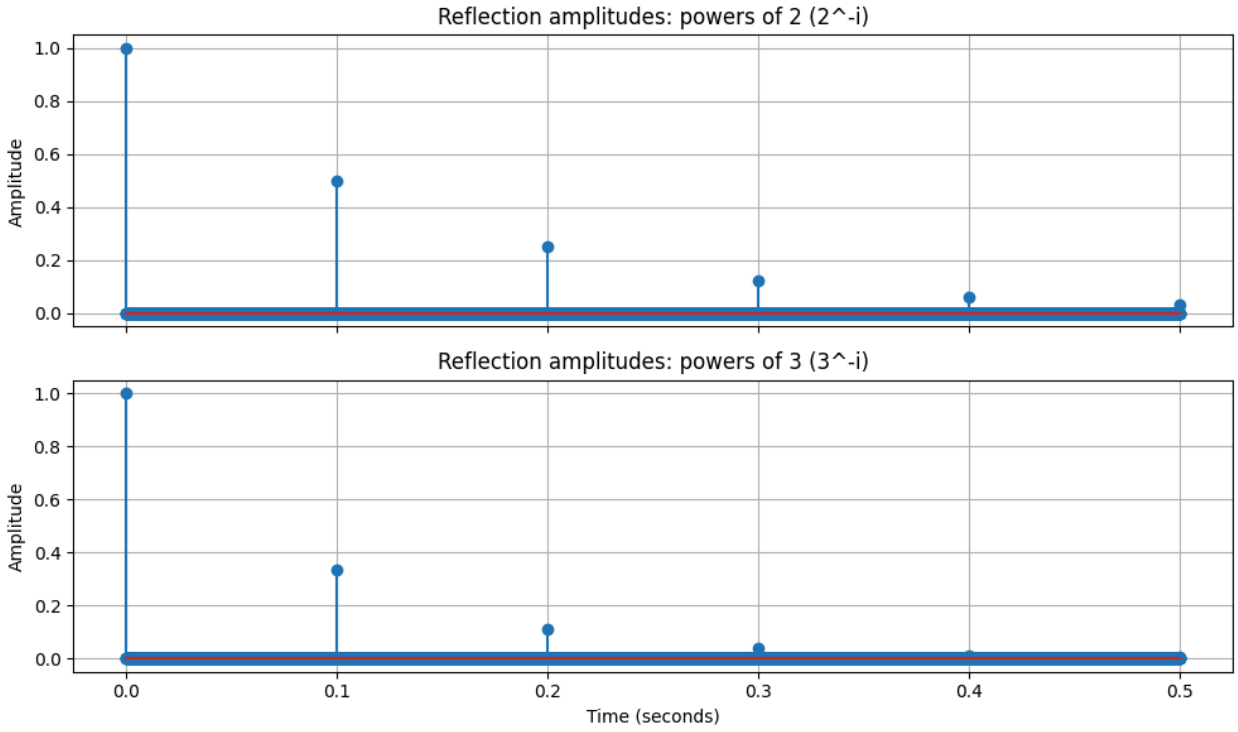
Frequency-Domain Analysis Using FFT

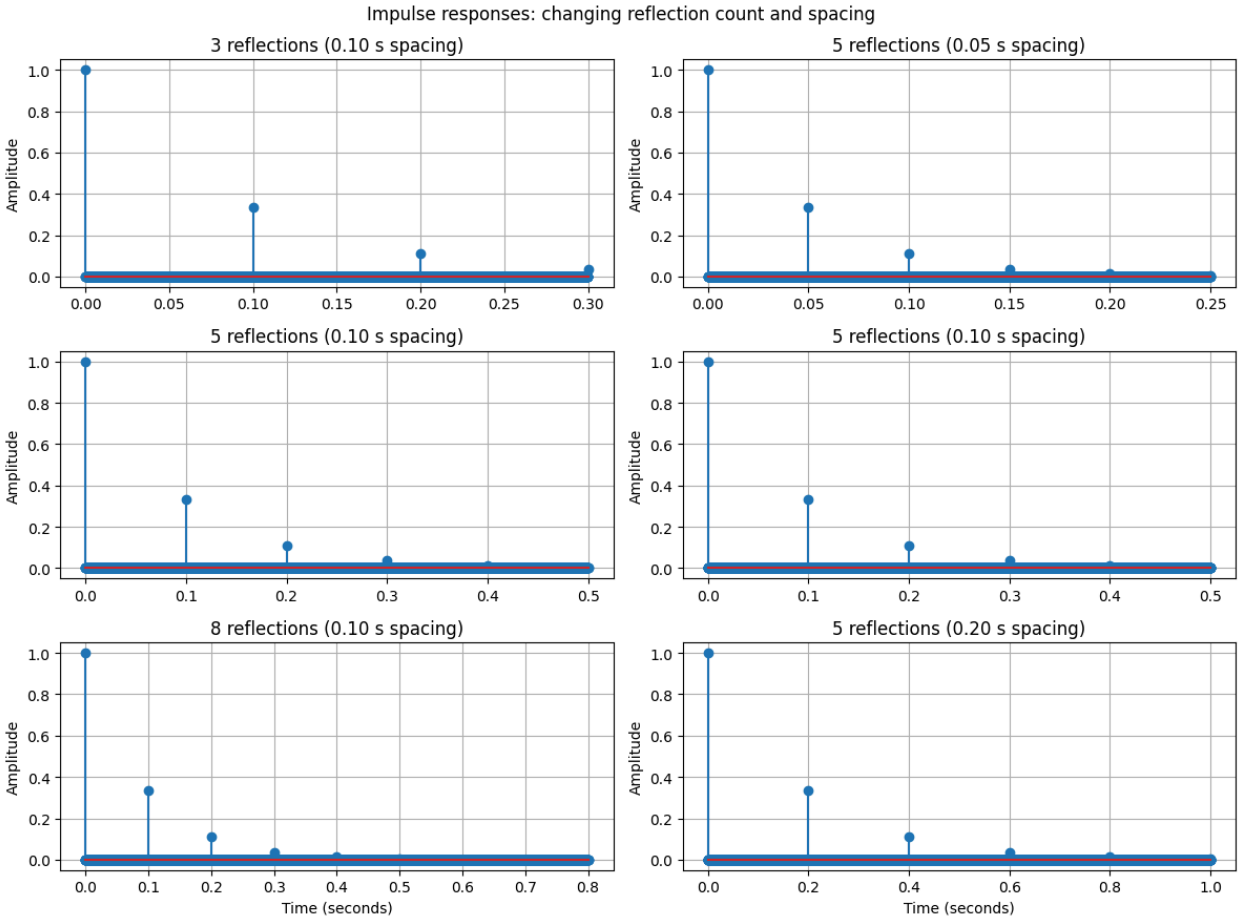
- To compare the original and processed audio in the frequency domain, we implemented the Discrete Fourier Transform using the radix-2 Cooley–Tukey FFT algorithm.
- The input samples were divided into two groups: even-indexed samples and odd-indexed samples.
- FFT was applied recursively to both groups until the sub-problems contained only one sample.

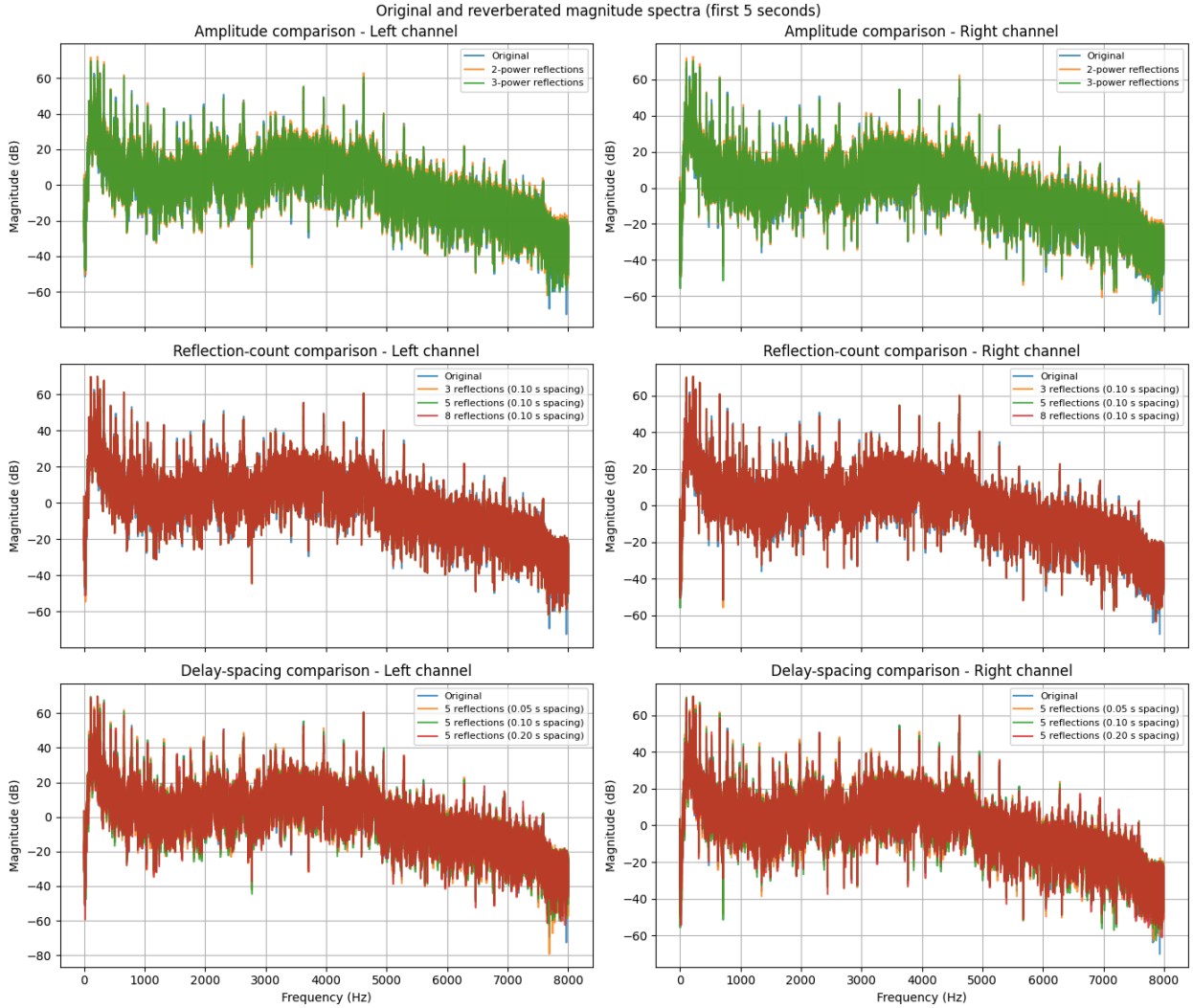
- The resulting frequency components were combined using complex twiddle factors, adding and subtracting the odd-part contributions to obtain the complete frequency spectrum.
- The FFT output was converted into a magnitude spectrum in decibels (dB), and only the non-negative frequency range was used for comparison.

3. Observations

- The powers-of-3 decay pattern produces weaker reflections than the powers-of-2 pattern, resulting in faster attenuation of later reflections.
- Increasing the number of reflections extends the duration of the reverberation effect.
- Smaller delay spacing places reflections closer together, while larger spacing spreads them further apart.
- Convolution introduces delayed copies of the original signal, changing its overall waveform and frequency content.
- The magnitude spectrum comparison shows how the reflection amplitudes and delay positions influence the frequency content of the processed audio.
- Overall, the experiment demonstrates how an impulse response can be used to simulate room reverberation and how changing its parameters affects the output sound.




Impulse responses: changing reflection count and spacing
3 reflections (0.10 s spacing)
5 reflections (0.05 s spacing)
5 reflections (0.10 s spacing)
5 reflections (0.10 s spacing)
8 reflections (0.10 s spacing)
5 reflections (0.20 s spacing)
Amplitude
Time (seconds)


Original and reverberated magnitude spectra (first 5 seconds)
Amplitude comparison - Left channel
Amplitude comparison - Right channel
Reflection-count comparison - Left channel
Reflection-count comparison - Right channel
Delay-spacing comparison - Left channel
Delay-spacing comparison - Right channel
Original
2-power reflections
3-power reflections
3 reflections (0.10 s spacing)
5 reflections (0.10 s spacing)
8 reflections (0.10 s spacing)
5 reflections (0.05 s spacing)
5 reflections (0.20 s spacing)
Magnitude (dB)
Frequency (Hz)

EXPERIMENT 2 – TIME-VARYING DELAY EFFECT

1. Objective

To implement a time-varying delay effect in an audio signal, where the delay changes with time, and compare its behaviour with a conventional fixed-delay echo.

2. Approach

Time-Varying Delay Generation

- We used the delay model:
  - $D(n) = D_0 + \Delta D \sin\left(\frac{2\pi f_m n}{f_s}\right)$
- The nominal delay (D0) was set to 20 ms, while the delay variation (Delta D) was set to 5 ms, so the delay changes approximately between 15 ms and 25 ms.
- The modulation frequency (fm) was set to 1 Hz, determining how quickly the delay varies over time.
- The calculated delay values were rounded to integer sample positions for accessing the audio samples.

Fixed-Delay and Time-Varying Audio Processing

- We first generated a fixed-delay echo using the nominal delay of 20 ms, with an echo scaling factor (alpha) of 0.5.
- For the time-varying effect, we calculated the source sample position for every output sample using (n-D(n)).
- Valid source positions were selected, and the corresponding delayed samples were multiplied by (alpha) and added to the original signal.
- The processing was performed independently on the left and right channels of the stereo recording.
- Both processed signals were saved as separate floating-point WAV files for comparison.

Time-Domain Analysis

- We plotted the original, fixed-delay and time-varying signals together for both stereo channels.
- The first 0.1 seconds of the recording were considered for waveform comparison.
- This helped us examine how the changing delay modifies the original waveform compared with a constant delay.

Frequency-Domain Analysis Using FFT

- We implemented the Discrete Fourier Transform using the radix-2 Cooley–Tukey FFT algorithm, with zero-padding to the next power of two when required.

- The input samples were divided into two groups: even-indexed samples and odd-indexed samples.
- FFT was applied recursively to both groups until the sub-problems contained only one sample.
- The resulting frequency components were combined using complex twiddle factors to obtain the complete frequency spectrum.
- The magnitude spectrum was calculated in decibels (dB), considering only the non-negative frequency range.
- The first 5 seconds of the original, fixed-delay and time-varying signals were compared for both stereo channels.

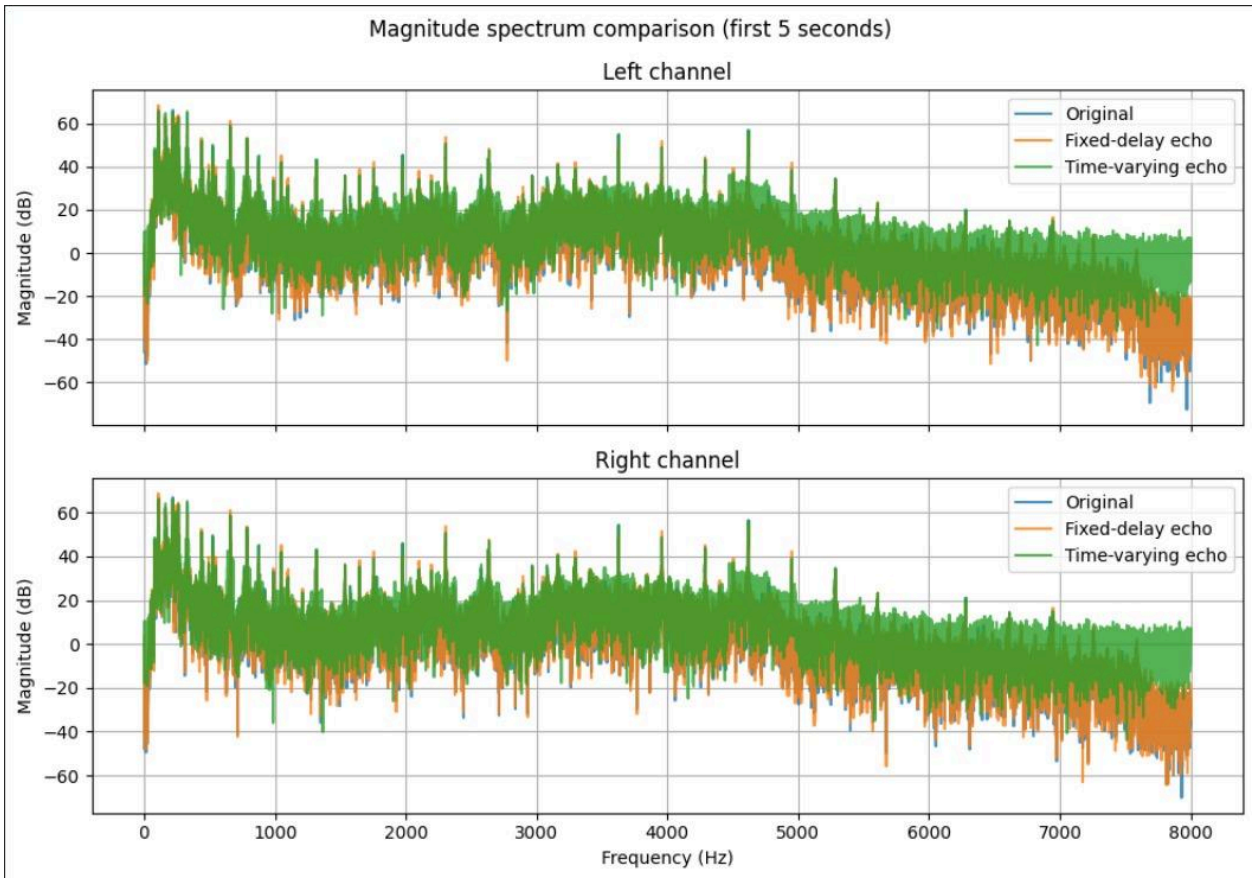
3. Observations

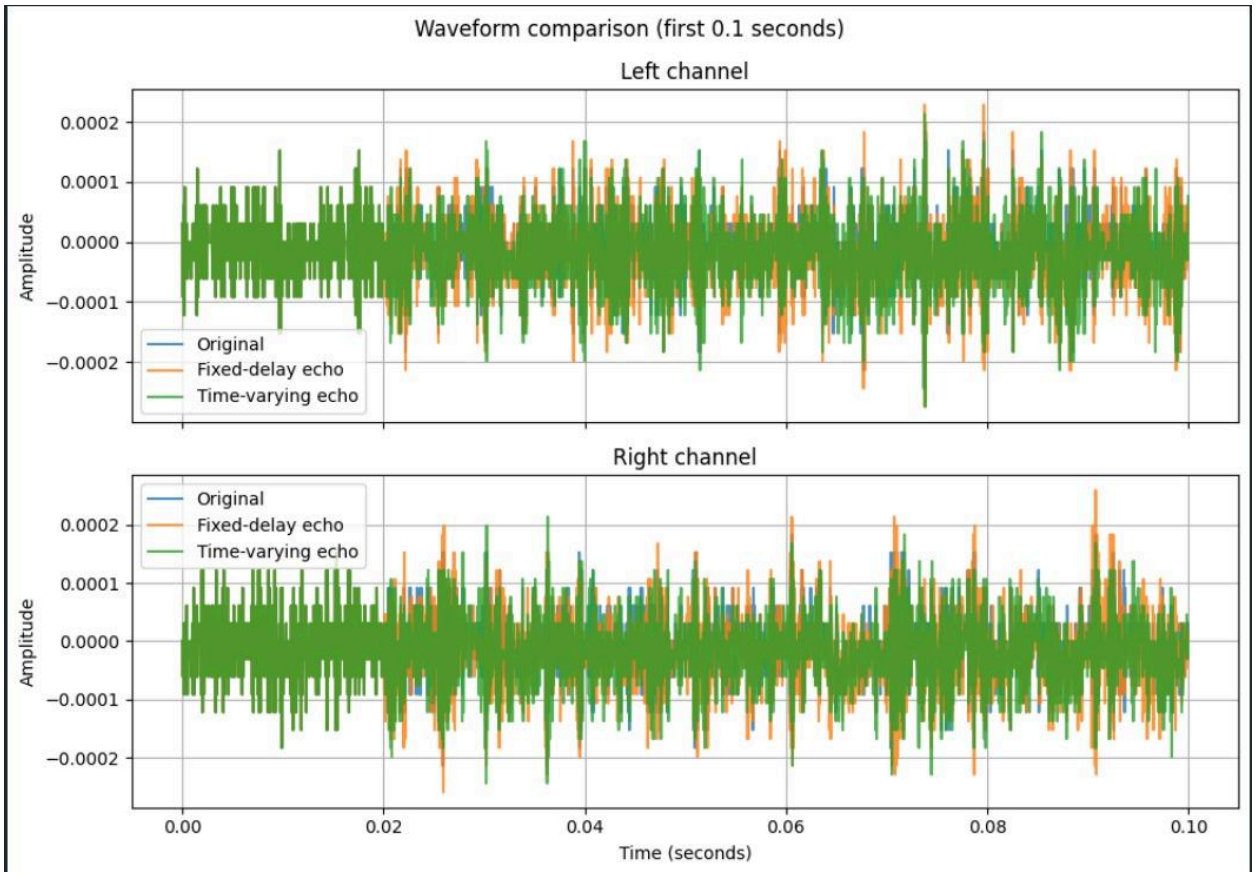
- In the fixed-delay case, the echo comes from 20 ms earlier in the original audio, whereas in the time-varying case, the delay changes between approximately 15 ms and 25 ms.
- The changing delay makes the position of the delayed audio vary over time, unlike the fixed-delay case.
- The waveform plots help us compare how fixed and time-varying delays change the original audio signal.
- The magnitude spectra help us compare the frequency content of the original audio and the two processed signals.
- The time-varying delay provides a changing echo effect, whereas the fixed-delay implementation produces an echo with a constant delay.
- Overall, the experiment demonstrates how a changing delay can be implemented using sinusoidal variation and studied in both the time and frequency domains.

Comparison with Experiment 1

Both experiments introduce delayed copies of the original audio. However, Experiment 1 uses multiple reflections with fixed spacing, whereas Experiment 2 uses a delay that varies continuously with time.

- Similar to Experiment 1, we added a delayed and scaled copy of the original audio signal to produce an echo effect.
- In Experiment 1, the spacing between consecutive reflections remains constant, whereas in Experiment 2, the delay changes sinusoidally with time.
- The fixed-delay output from Experiment 2 was used as a reference to compare the effect of the time-varying delay.
- This comparison helps us understand how changing the delay over time affects the audio signal compared with a constant delay.


Magnitude spectrum comparison (first 5 seconds)
Left channel
Right channel
Original
Fixed-delay echo
Time-varying echo
Magnitude (dB)
Frequency (Hz)
60
40
20
0
−20
−40
−60
0
1000
2000
3000
4000
5000
6000
7000
8000


Waveform comparison (first 0.1 seconds)
Left channel
Right channel
Amplitude
Time (seconds)
Original
Fixed-delay echo
Time-varying echo
0.0002
0.0001
0.0000
−0.0001
−0.0002
0.00
0.02
0.04
0.06
0.08
0.10